ASSIGNMENT 3: GIT & R

Sakila Database Analysis

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GIT Repo: https://github.com/mahmad289/MSDS25061_Assignment_3

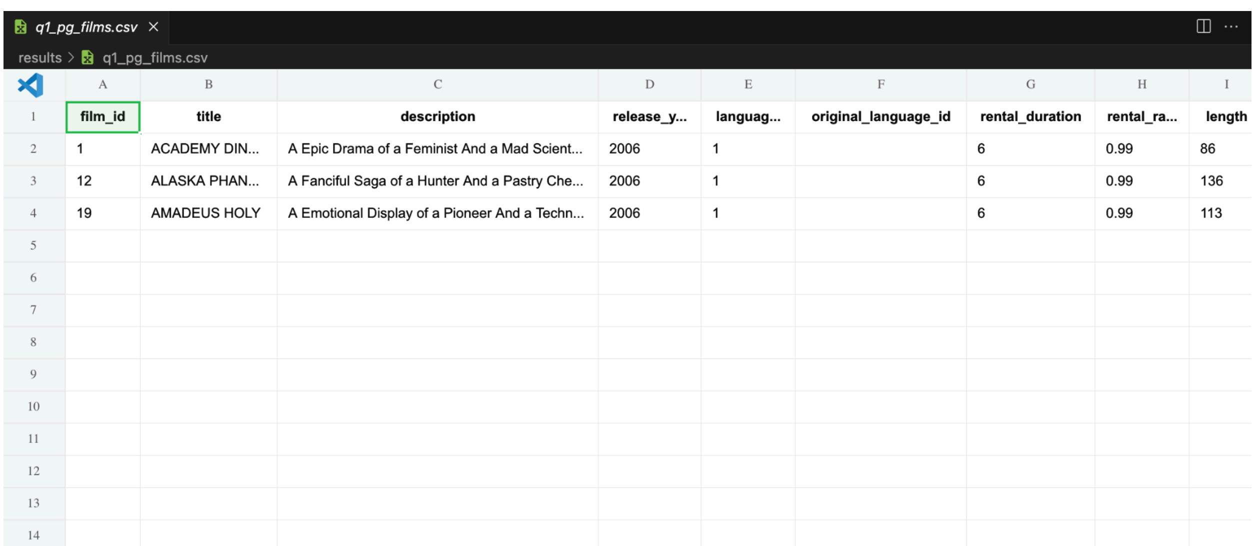
Question 1: Films with PG Rating and Rental Duration > 5 Days

Code:

```
pg_long_rental_films <- films_dt[rating == "PG" & rental_duration > 5]
fwrite(pg_long_rental_films, "results/q1_pg_films.csv")
```

Explanation: This filters the films table for movies rated PG with rental duration greater than 5 days using data.table syntax.

Screenshot:



q1_pg_films.csv
results > q1_pg_films.csv

film_id	title	description	release_y...	languag...	original_language_id	rental_duration	rental_ra...	length
1	ACADEMY DIN...	A Epic Drama of a Feminist And a Mad Scient...	2006	1		6	0.99	86
12	ALASKA PHAN...	A Fanciful Saga of a Hunter And a Pastry Che...	2006	1		6	0.99	136
19	AMADEUS HOLY	A Emotional Display of a Pioneer And a Techn...	2006	1		6	0.99	113

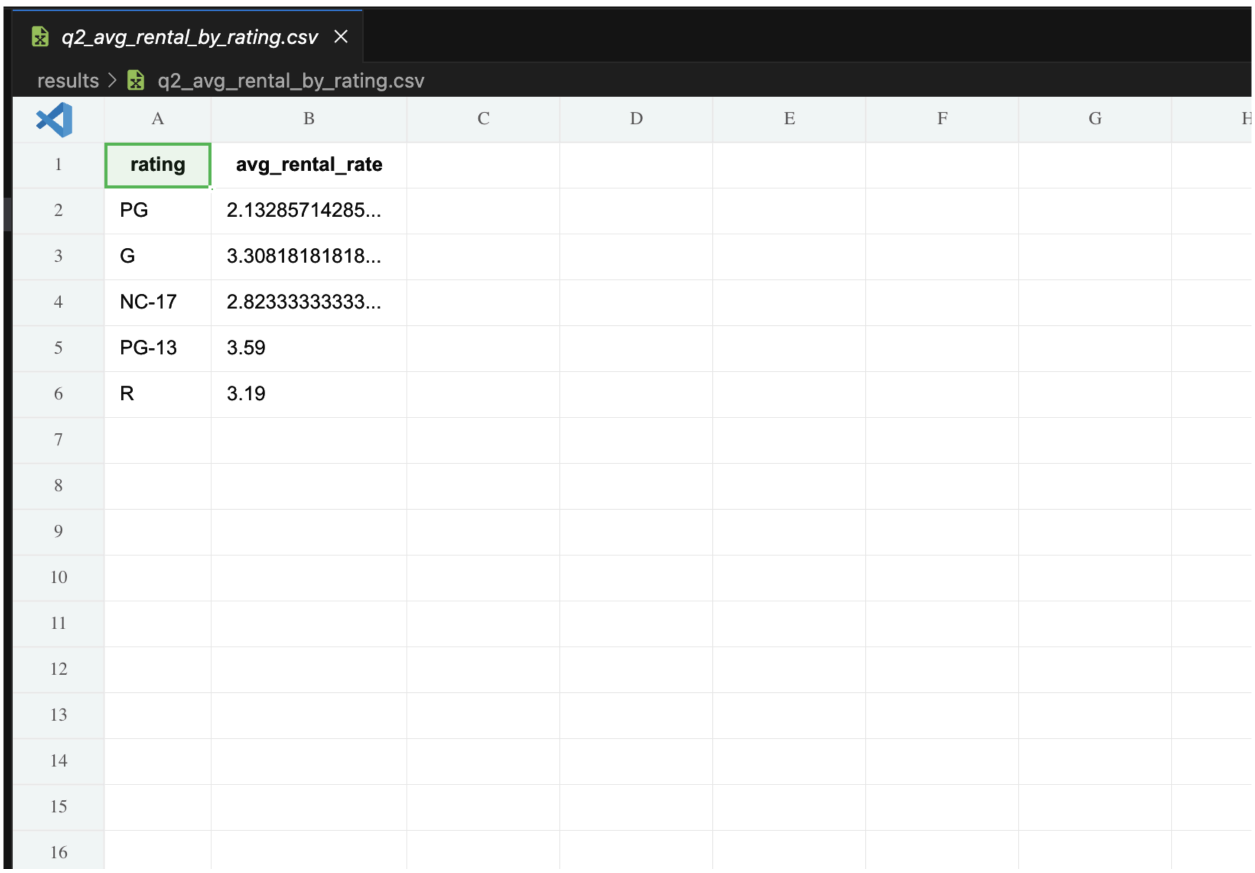
Question 2: Average Rental Rate by Film Rating

Code:

```
avg_rental_rate_by_rating <- films_dt[, .(avg_rental_rate = mean(rental_rate, na.rm = TRUE)),
by = rating]
fwrite(avg_rental_rate_by_rating, "results/q2_avg_rental_by_rating.csv")
```

Explanation: This calculates the mean rental rate for each rating category using data.table's grouping functionality.

Screenshot: *Output showing average rental rates by rating*



rating	avg_rental_rate
PG	2.13285714285...
G	3.30818181818...
NC-17	2.82333333333...
PG-13	3.59
R	3.19

Question 3: Film Count by Language

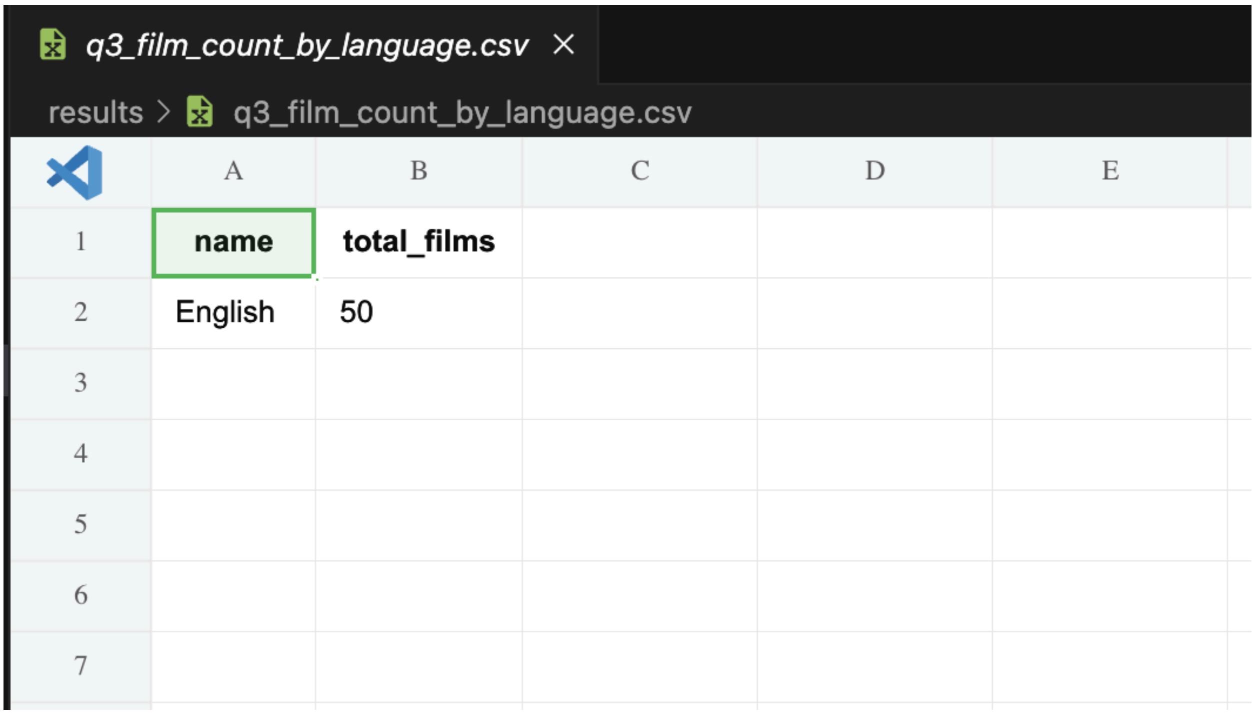
Code:

```
films_with_language <- merge(films_dt, languages_dt, by.x = "language_id", by.y =
"language_id")
film_counts_by_language <- films_with_language[, .N, by = name]
```

```
setnames(film_counts_by_language, "N", "total_films")
fwrite(film_counts_by_language, "results/q3_film_count_by_language.csv")
```

Explanation: This joins the films and languages tables, then counts films per language using .N to count rows in each group.

Screenshot: *Film counts by language output*



	A	B	C	D	E
1	name	total_films			
2	English	50			
3					
4					
5					
6					
7					

Question 4: Customer and Store Information

Code:

```
customers_with_store <- merge(customers_dt, stores_dt, by.x = "store_id", by.y = "store_id")
customers_store_list <- customers_with_store[, .(first_name, last_name, store_id)]
fwrite(customers_store_list, "results/q4_customers_store.csv")
```

Explanation: This merges customer and store tables to list customer names with their associated store IDs.

Screenshot: *Customer and store list output* q4_customers_store.csv

results > q4_customers_store.csv

	A	B	C	D	E	F
1	**first_na...**	**last_n...**	**store...**			
2	MARY	SMITH	1			
3	PATRIC...	JOHN...	1			
4	LINDA	WILLI...	1			
5	ELIZAB...	BROWN	1			
6	MARIA	MILLER	1			
7	DOROT...	TAYL...	1			
8	NANCY	THOM...	1			
9	HELEN	HARRIS	1			
10	DONNA	THOM...	1			
11	RUTH	MART...	1			
12	MICHE...	CLARK	1			
13	LAURA	RODR...	1			
14	DEBOR...	WALK...	1			
15	CYNTHIA	YOUNG	1			
16	MELISSA	KING	1			
17	AMY	LOPEZ	1			
18	PAMELA	BAKER	1			

Question 5: Payment Details with Staff Information

Code:

```
payments_with_staff <- merge(payments_dt, staff_dt, by.x = "staff_id", by.y = "staff_id")
payment_records_with_staff <- payments_with_staff[, .(
  amount,
  payment_date,
  staff_first_name = first_name,
  staff_last_name = last_name
)]
fwrite(payment_records_with_staff, "results/q5_payment_staff.csv")
```

Explanation: This joins payment and staff tables to show payment amount, date, and staff member who processed it.

Screenshot: *Payment records with staff details*

q5_payment_staff.csv

results > q5_payment_staff.csv

	A	B	C	D	E	F
1	amount	payment_date	staff_first_name	staff_last_na...		
2	2.99	2005-05-25T11:30:37Z	Mike	Hillyer		
3	0.99	2005-05-28T10:35:23Z	Mike	Hillyer		
4	5.99	2005-06-15T00:54:12Z	Mike	Hillyer		
5	4.99	2005-06-16T15:18:57Z	Mike	Hillyer		
6	4.99	2005-06-18T08:41:48Z	Mike	Hillyer		
7	3.99	2005-06-21T06:24:45Z	Mike	Hillyer		
8	5.99	2005-07-08T07:33:56Z	Mike	Hillyer		
9	4.99	2005-07-09T13:24:07Z	Mike	Hillyer		
10	4.99	2005-07-09T16:38:01Z	Mike	Hillyer		
11	7.99	2005-07-11T10:13:46Z	Mike	Hillyer		
12	4.99	2005-07-28T09:04:45Z	Mike	Hillyer		
13	0.99	2005-07-28T17:33:39Z	Mike	Hillyer		
14	0.99	2005-08-02T18:01:38Z	Mike	Hillyer		
15	0.99	2005-08-18T03:57:29Z	Mike	Hillyer		
16	0.99	2005-08-21T23:33:57Z	Mike	Hillyer		
17	1.99	2005-08-22T01:27:57Z	Mike	Hillyer		
18	5.99	2005-08-22T20:03:46Z	Mike	Hillyer		

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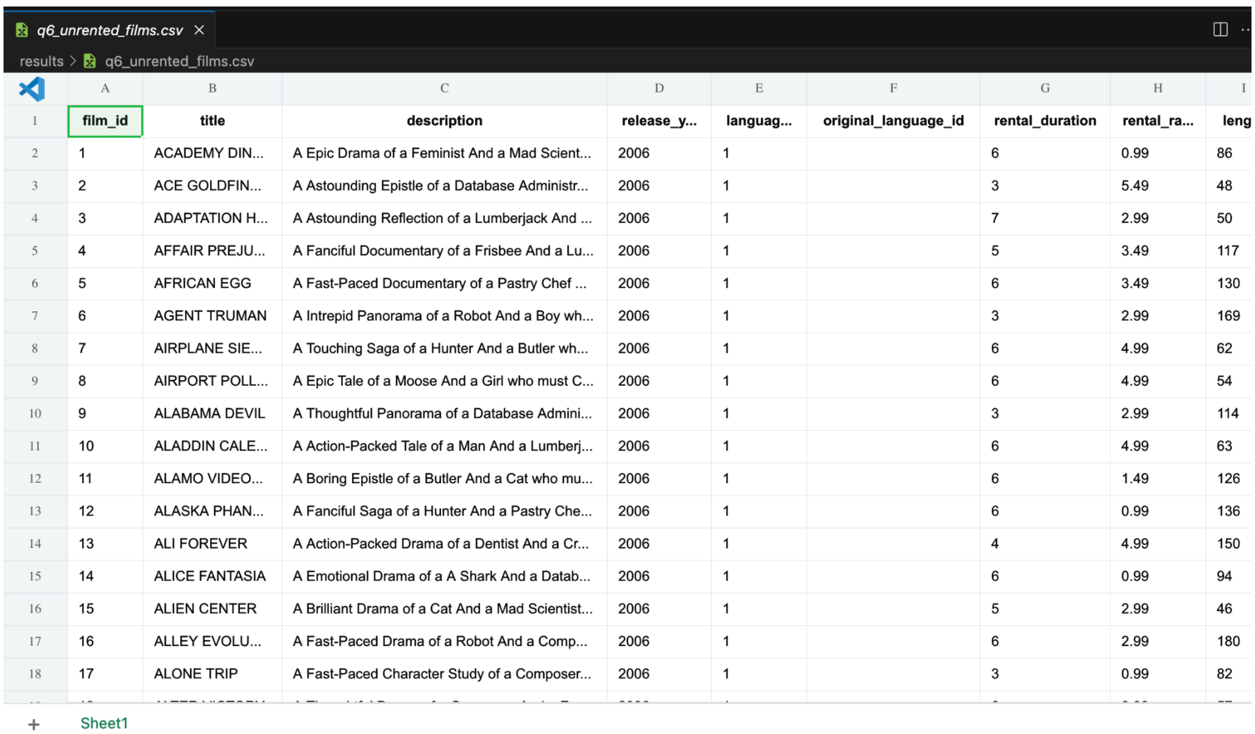
Question 6: Films Not Rented

Code:

```
rented_film_ids <- unique(rentals_dt$film_id)
films_never_rented <- films_dt[!(film_id %in% rented_film_ids)]
fwrite(films_never_rented, "results/q6_unrented_films.csv")
```

Explanation: This identifies films that never appear in the rentals table by using an anti-join pattern with the %in% operator.

Screenshot: *List of unrented films*



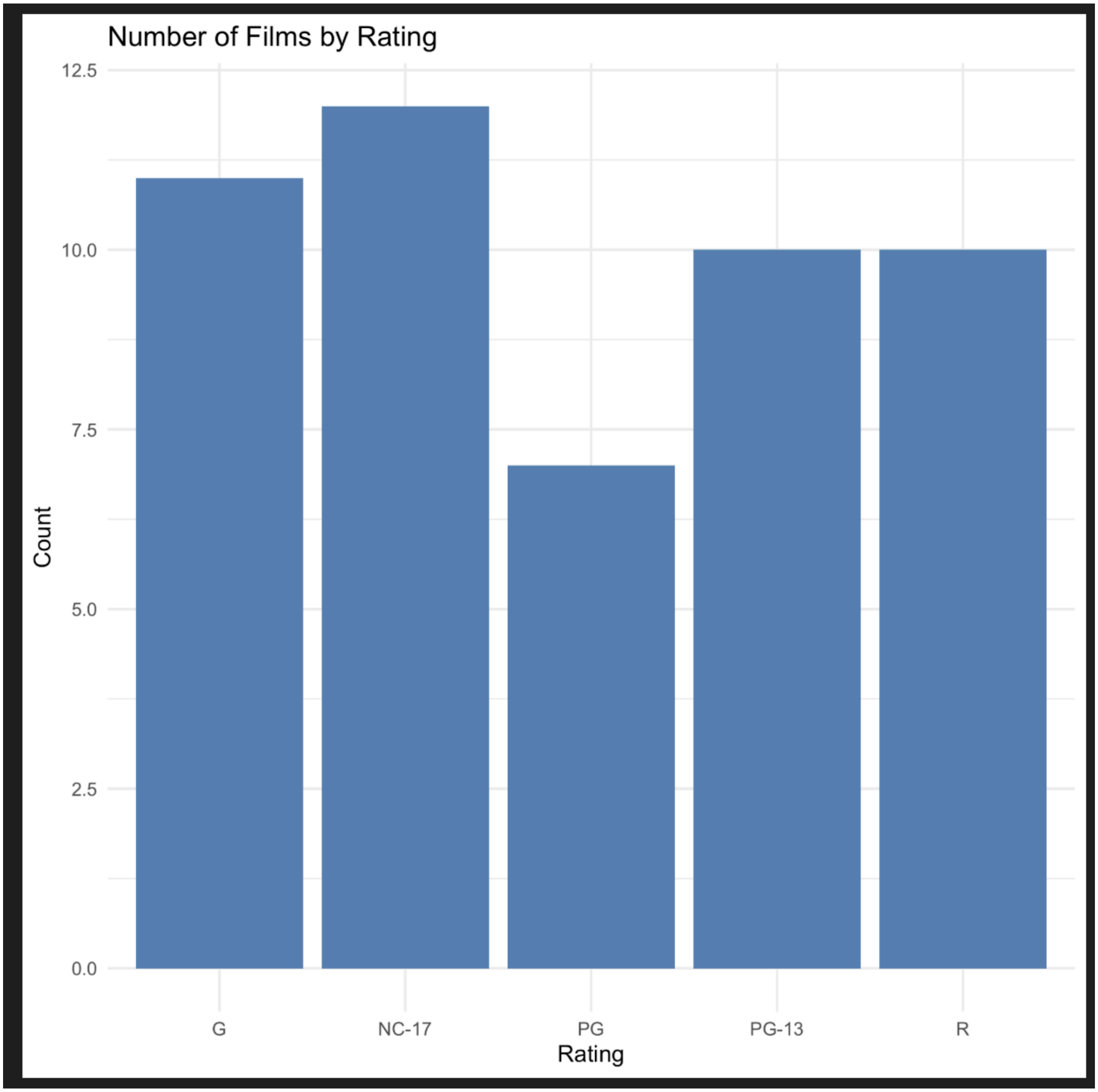
film_id	title	description	release_y...	languag...	original_language_id	rental_duration	rental_ra...	leng
1	ACADEMY DIN...	A Epic Drama of a Feminist And a Mad Scient...	2006	1		6	0.99	86
2	ACE GOLDFIN...	A Astounding Epistle of a Database Administr...	2006	1		3	5.49	48
3	ADAPTATION H...	A Astounding Reflection of a Lumberjack And ...	2006	1		7	2.99	50
4	AFFAIR PREJU...	A Fanciful Documentary of a Frisbee And a Lu...	2006	1		5	3.49	117
5	AFRICAN EGG	A Fast-Paced Documentary of a Pastry Chef ...	2006	1		6	3.49	130
6	AGENT TRUMAN	A Intrepid Panorama of a Robot And a Boy wh...	2006	1		3	2.99	169
7	AIRPLANE SIE...	A Touching Saga of a Hunter And a Butler wh...	2006	1		6	4.99	62
8	AIRPORT POLL...	A Epic Tale of a Moose And a Girl who must C...	2006	1		6	4.99	54
9	ALABAMA DEVIL	A Thoughtful Panorama of a Database Admini...	2006	1		3	2.99	114
10	ALADDIN CALE...	A Action-Packed Tale of a Man And a Lumberj...	2006	1		6	4.99	63
11	ALAMO VIDEO...	A Boring Epistle of a Butler And a Cat who mu...	2006	1		6	1.49	126
12	ALASKA PHAN...	A Fanciful Saga of a Hunter And a Pastry Che...	2006	1		6	0.99	136
13	ALI FOREVER	A Action-Packed Drama of a Dentist And a Cr...	2006	1		4	4.99	150
14	ALICE FANTASIA	A Emotional Drama of a A Shark And a Datab...	2006	1		6	0.99	94
15	ALIEN CENTER	A Brilliant Drama of a Cat And a Mad Scientist...	2006	1		5	2.99	46
16	ALLEY EVOLU...	A Fast-Paced Drama of a Robot And a Comp...	2006	1		6	2.99	180
17	ALONE TRIP	A Fast-Paced Character Study of a Composer...	2006	1		3	0.99	82

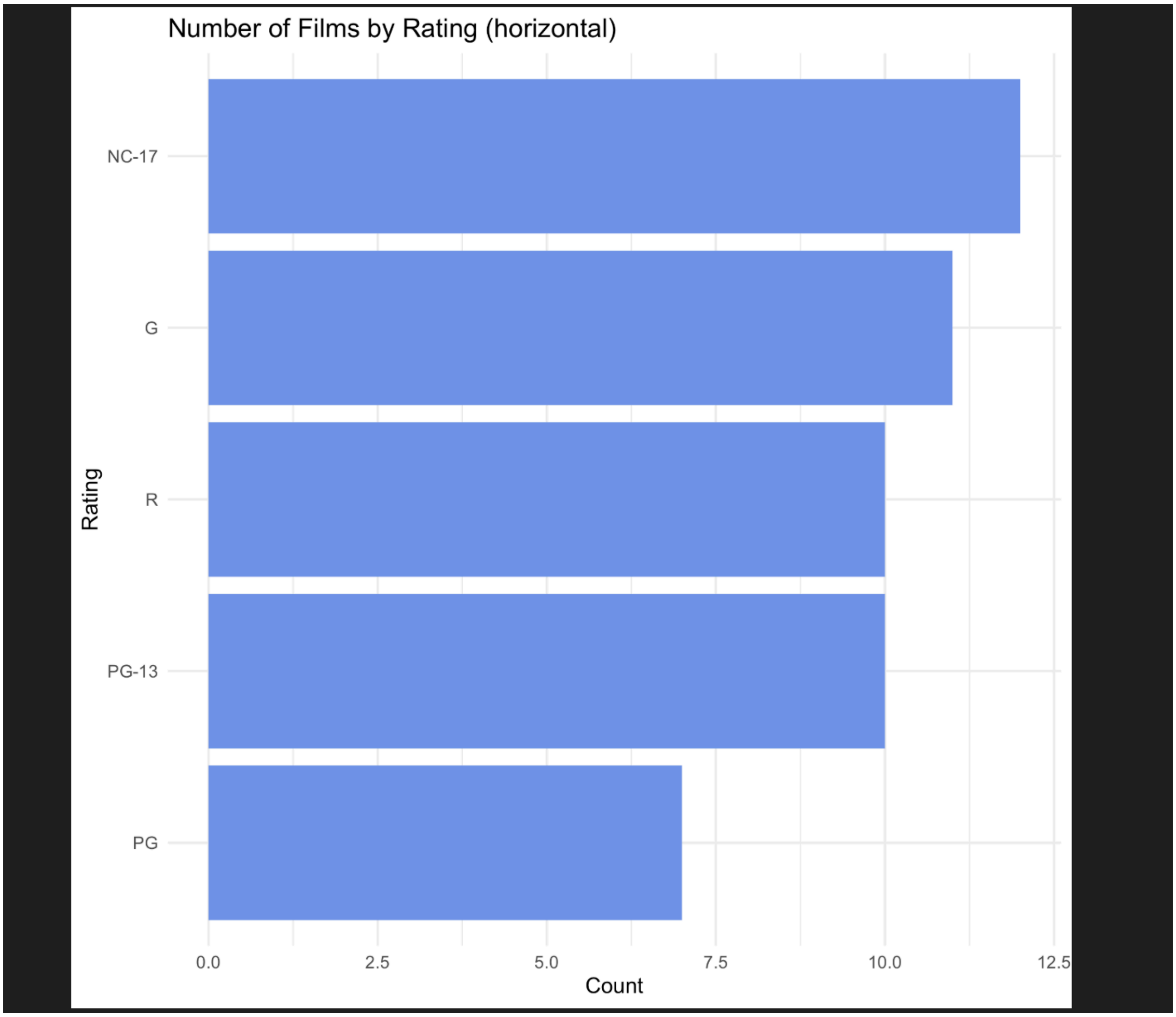
Question 7: Visualizations

7.1 Films by Rating (Bar Chart)

```
films_by_rating_plot <- ggplot(films_dt, aes(x = rating)) +
  geom_bar(fill = "steelblue") +
  theme_minimal() +
  labs(title = "Number of Films by Rating", x = "Rating", y = "Count")
ggsave("visualizations/q7_films_by_rating.png", plot = films_by_rating_plot)
```

Screenshot: *Bar chart showing films by rating*





7.2 Average Rental Rate by Rating

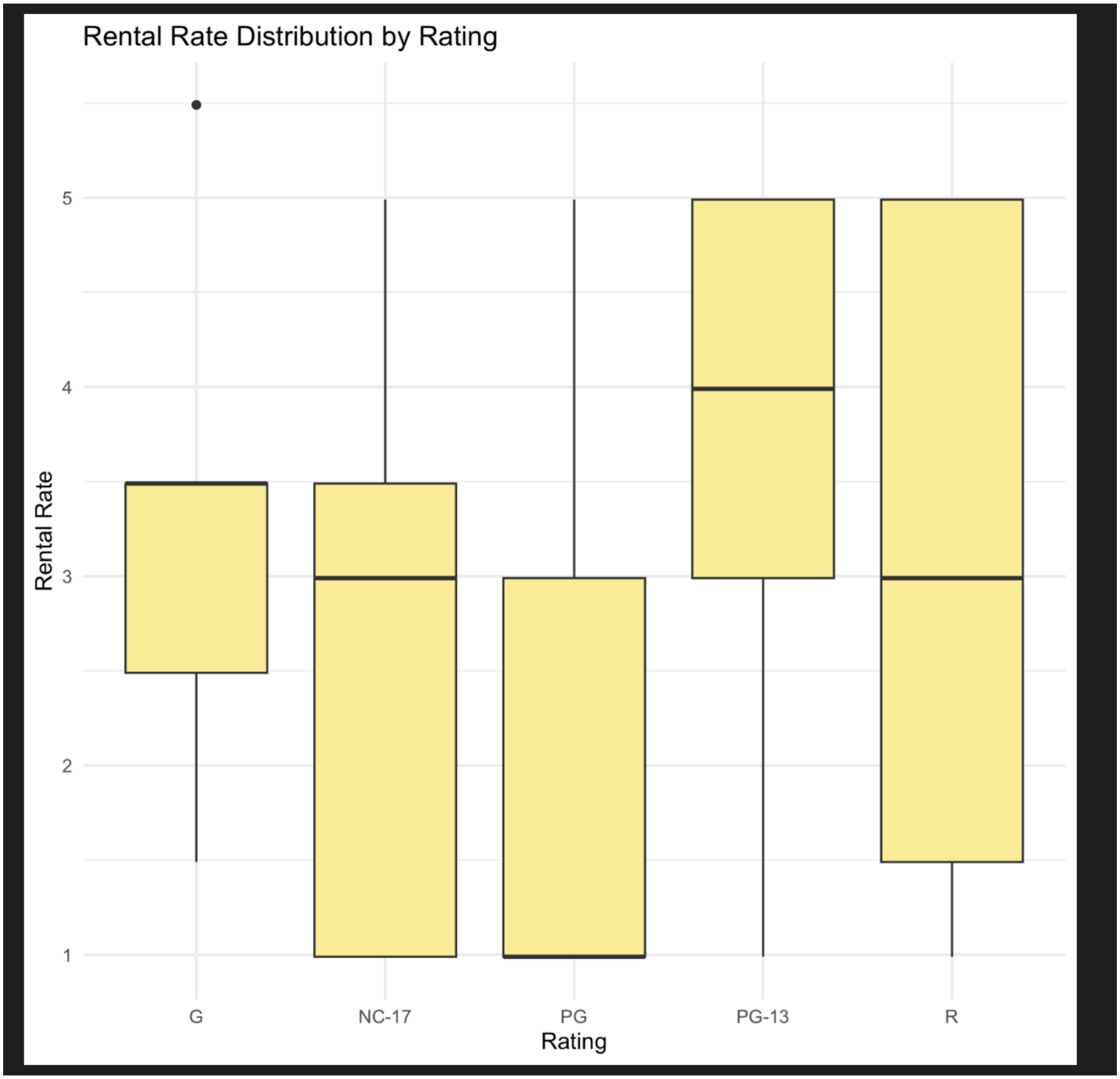
```
films_by_rating_avg_rate <- ggplot(avg_rental_rate_by_rating, aes(x = rating, y =
avg_rental_rate)) +
  geom_col(fill = "seagreen3") +
  theme_minimal() +
  labs(title = "Average Rental Rate by Rating", x = "Rating", y = "Average Rental Rate")
ggsave("visualizations/q7c_avg_rental_rate_by_rating.png", plot = films_by_rating_avg_rate)
```

Screenshot: *Bar chart showing average rental rates*

7.3 Rental Rate Distribution (Box Plot)

```
films_rental_rate_boxplot <- ggplot(films_dt, aes(x = rating, y = rental_rate)) +
  geom_boxplot(fill = "lightgoldenrod1") +
  theme_minimal() +
  labs(title = "Rental Rate Distribution by Rating", x = "Rating", y = "Rental Rate")
ggsave("visualizations/q7d_rental_rate_boxplot_by_rating.png", plot =
films_rental_rate_boxplot)
```

Screenshot: *Box plot showing rental rate distribution*



Explanation: Three different visualizations were created to analyze films by rating and rental rates, showing distribution and pricing patterns.

7.4 Rental Rate Distribution (Pie)

```
counts_by_rating[, pct := count / sum(count) * 100]
films_by_rating_pie <- ggplot(counts_by_rating, aes(x = "", y = count, fill = rating)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  theme_void() +
  labs(title = "Share of Films by Rating") +
```

```
  geom_text(aes(label = paste0(round(pct, 1), "%")), position = position_stack(vjust = 0.5), color
= "white", size = 3)
ggsave("visualizations/q7b_films_by_rating_pie.png", plot = films_by_rating_pie)
```

Screenshot: *Box plot showing rental rate distribution*